

HEART ATTACK PREDICTION USING DEEP LEARNING

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SEMINAR DOMAIN: Deep Learning

Deep learning is the subset of machine learning methods which are based on artificial neural networks with representation learning. The adjective "deep" in deep learning refers to the use of multiple layers in the network. Methods used can be either supervised, semi-supervised or unsupervised.

Deep-learning architectures such as deep neural networks, deep belief networks, deep reinforcement learning, recurrent neural networks, convolutional neural networks and transformers have been applied to fields including computer vision, speech recognition, natural language processing, machine translation, bioinformatics, drug design, medical image analysis, climate science, material inspection and board game programs, where they have produced results comparable to and in some cases surpassing human expert performance.

Artificial neural networks (ANNs) were inspired by information processing and distributed communication nodes in biological systems. ANNs have various differences from biological brains. Specifically, artificial neural networks tend to be static and symbolic, while the biological brain of most living organisms is dynamic (plastic) and analog.

REFERENCES: https://en.wikipedia.org/wiki/Deep_learning

SEMINAR SUB DOMAIN: Heart Attack Prediction Using Deep Learning

Cardiovascular diseases state as one of the greatest risks of death for the general population. Late detection in heart diseases highly conditions the chances of survival for patients. Age, sex, cholesterol level, sugar level, heart rate, among other factors, are known to have an influence on life-threatening heart problems, but, due to the high amount of variables, it is often difficult for an expert to evaluate each patient taking this information into account. In this manuscript, the authors propose using deep learning methods, combined with feature augmentation techniques for evaluating whether patients are at risk of suffering cardiovascular disease. The results of the proposed methods outperform other state of the art

methods by 4.4%, leading to a precision of a 90%, which presents a significant improvement, even more so when it comes to an affliction that affects a large population.

REFERENCES: <https://link.springer.com/article/10.1007>

SEMINAR TOPIC :Heart Attack Prediction Using Deep Learning

Cardiovascular disease, including silent heart attacks, is a serious and often hard-to-diagnose health issue. The lack of specialist doctors and the high rate of misdiagnosis have created a pressing need for an effective system to detect heart disease. While various medical data mining and machine learning techniques have been attempted to predict heart disease, their accuracy has fallen short.

This paper proposes a solution involving deep learning, specifically Recurrent Neural Networks (RNNs), to predict the likelihood of heart-related diseases. RNNs are powerful tools within artificial neural networks for classification. The paper aims to detail the key components of this system and the associated theory. The proposed model combines deep learning and data mining to deliver accurate results with minimal errors, offering a more reliable approach to heart attack prediction.

The paper also serves as a guide and precedent for the development of a new generation of heart attack prediction platforms. In essence, it seeks to advance the field of cardiovascular disease prediction by harnessing the potential of deep learning and RNNs.

REVIEW PAPER-I: Heart Attack Prediction Using Deep Learning

Cardiovascular disease is one of the most heinous disease, especially the silent heart attack, which attacks a person so abruptly that there's no time to get it treated and such disease is very difficult to be diagnosed. The lack of specialist doctors and increase in wrong diagnosed cases has necessitated the need for building an efficient heart disease detection system. Various medical data mining and machine learning techniques are being implemented to extract the valuable information regarding the heart disease prediction. Yet, the accuracy of the desired results are not satisfactory. This paper proposes a heart attack prediction system using Deep learning techniques, specifically Recurrent Neural Network to predict the likely possibilities of heart related diseases of the patient. Recurrent Neural Network is a very powerful classification algorithm that makes use of Deep Learning approach in Artificial Neural Network. The paper discusses in detail the major modules of the system along with the related theory. The proposed model incorporates deep learning and data mining to provide the

accurate results with minimum errors. This paper provides a direction and precedent for the development of a new breed of heart attack prediction platform.

ALGORITHM USED:

1. Recurrent Neural Network (RNN): The paper employs RNNs, a type of artificial neural network designed for sequential data analysis. RNNs are well-suited for tasks involving time series data, making them a valuable choice for heart attack prediction, which often relies on the analysis of patient data over time.
2. Gated Recurrent Unit (GRU): Within the context of RNNs, the paper specifically uses the Gated Recurrent Unit (GRU). GRU is a type of RNN architecture that incorporates gating mechanisms to address the vanishing gradient problem. It allows for the modeling of dependencies in sequences and is applied to the heart attack prediction task.

REFERENCE:[Kishore, A., Kumar, A., Singh, K., Punia, M., & Hambir, Y. \(2018\). Heart Attack Prediction Using Deep Learning. International Research Journal of Engineering and Technology \(IRJET\), 5\(4\), 4420-4423.](#)

REVIEW PAPER-2: Prediction of Heart Disease using Machine Learning Algorithms: A Survey

According to recent survey by WHO organisation 17.5 million people dead each year. It will increase to 75 million in the year 2030[1]. Medical professionals working in the field of heart disease have their own limitation, they can predict chance of heart attack up to 67% accuracy[2], with the current epidemic scenario doctors need a support system for more accurate prediction of heart disease. Machine learning algorithm and deep learning opens new door opportunities for precise predication of heart attack. Paper provideslot information about state of art methods in Machine learning and deep learning. An analytical comparison has been provided to help new researches' working in this field

ALGORITHM USED: ANNs are a broad category of machine learning models that consist of multiple layers of interconnected nodes (neurons) and are widely used for various tasks, including prediction and classification. The exact architecture and parameters of an ANN would depend on the specific implementation and dataset used for the heart disease prediction. Different types of ANN architectures, such as feedforward neural networks or recurrent neural networks (RNNs), can be used, depending on the nature of the data and the problem.

REFERENCE: Bharti, R., Khamparia, A., Shabaz, M., Dhiman, G., Pande, S., & Singh, P. (2021). Prediction of heart disease using a combination of

machine learning and deep learning. Computational intelligence and neuroscience, 2021.

REVIEW PAPER-3: Cardiovascular disease prediction using deep learning techniques

Cardiovascular Disease or coronary illness is one of the significant dangerous infections in India as well as in the entire world. It is estimated that 28.1 % of deaths occur due to heart diseases. It is also the major cause for significant number of deaths which as more than 17.6 million in the year 2016. So proper and timely diagnosis, treatment of such diseases require a system that can predict with precise accuracy and reliability. Intensive research is carried out by various researchers using diverse machine learning algorithms to forecast the heart disease taking different datasets which consists of different attributes that result in heart attack. In this paper we analyzed the dataset collected from kaggle which consists of attributes related to heart disease such as age, gender, blood pressure, cholesterol and so on. We have also investigated the accuracy levels of various machine learning techniques such as Support Vector Machines (SVM), K-Nearest Neighbor (KNN), Decision Trees (DT). The performance and accuracy of above algorithms is not so well when executed using large dataset, so here we tried to improving the prediction accuracy using Artificial Neural Network (ANN), Tensor Flow Keras.

ALGORITHM USED IN THIS PAPER: Artificial Neural Networks (ANN) are a class of machine learning algorithms inspired by the structure and function of the human brain. ANNs consist of interconnected nodes, which are modeled after neurons in the brain. These nodes are organized into layers, typically consisting of an input layer, one or more hidden layers, and an output layer. The most common ANN algorithm involves feedforward neural networks, backpropagation, and various activation functions

REFERENCE: Pasha, S. N., Ramesh, D., Mohmmad, S., & Harshavardhan, A. (2020, December). Cardiovascular disease prediction using deep learning techniques. In IOP conference series: materials science and engineering (Vol. 981, No. 2, p. 022006). IOP Publishing.

REVIEW PAPER-4: An Efficient SMOTE-Based Deep Learning Model for Heart Attack Prediction

Cardiac disease treatments are often being subjected to the acquisition and analysis of vast quantity of digital cardiac data. ese data can be utilized for various beneficial purposes. ,ese data's utilization becomes more important when

we are dealing with critical diseases like a heart attack where patient life is often at stake. Machine learning and deep learning are two famous techniques that are helping in making the raw data useful. Some of the biggest problems that arise from the usage of the aforementioned techniques are massive resource utilization, extensive data preprocessing, need for features engineering, and ensuring reliability in classification results. ,e proposed research work presents a cost-effective solution to predict heart attack with high accuracy and reliability. It uses a UCI dataset to predict the heart attack via various machine learning algorithms without the involvement of any feature engineering. Moreover, the given dataset has an unequal distribution of positive and negative classes which can reduce performance. ,e proposed work uses a synthetic minority oversampling technique (SMOTE) to handle given imbalance data. ,e proposed system discarded the need of feature engineering for the classification of the given dataset.,is led to an efficient solution as feature engineering often proves to be a costly process.,e results show that among all machine learning algorithms, SMOTE-based artificial neural network when tuned properly outperformed all other models and many existing systems. ,e high reliability of the proposed system ensures that it can be effectively used in the prediction of the heart attack.

ALGORITHM USED: ANNs are a broad category of machine learning models that consist of multiple layers of interconnected nodes (neurons) and are widely used for various tasks, including prediction and classification. The exact architecture and parameters of an ANN would depend on the specific implementation and dataset used for the heart disease prediction. Different types of ANN architectures, such as feedforward neural networks or recurrent neural networks (RNNs), can be used, depending on the nature of the data and the problem.

REFERENCE: Waqar, M., Dawood, H., Dawood, H., Majeed, N., Banjar, A., & Alharbey, R. (2021). An efficient SMOTE-based deep learning model for heart attack prediction. Scientific Programming, 2021, 1-12.

FUTURE WORK/PLAN

Recurrent Neural Network (RNN): The paper employs RNNs, a type of artificial neural network designed for sequential data analysis. RNNs are well-suited for tasks involving time series data, making them a valuable choice for heart attack prediction, which often relies on the analysis of patient data over time.

REFERENCE : <u>Kishore, A., Kumar, A., Singh, K., Punia, M., & Hambir, Y. (2018). Heart Attack Prediction Using Deep Learning. International Research Journal of Engineering and Technology (IRJET), 5(4), 4420-4423.</u>
